

ST CLOUD, MN, USA

WMO: 726550

Lat: 45.543N

Lon: 94.051W

Elev: 310

StdP: 97.65

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-24.1	-30.6	0.2	-26.4	-28.0	0.3	-23.4	11.8	-11.2	10.8	-10.4	2.4	310	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.9	22.6	30.2	21.5	28.6	20.5	24.4	29.5	23.3	28.2	22.1	26.7	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.2	27.7	21.6	16.9	26.1	20.5	15.8	25.0	75.5	29.2	70.7	28.3	66.4	26.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.1	8.2	DB	-31.3	34.8	3.7	1.5	-34.0	35.9	-36.1	36.8	-38.2	37.6	-40.9	38.7	(4)
(5)			WB	-30.8	26.4	2.5	1.2	-32.6	27.3	-34.0	28.0	-35.5	28.7	-37.3	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-11.1	-8.8	-1.4	6.5	13.5	19.1	21.6	20.1	15.8	8.2	-0.1	-7.6	(6)	
	DBStd	12.67	7.53	7.11	6.99	5.55	4.69	3.73	3.19	3.08	4.70	5.04	5.91	6.75	(7)	
	HDD10.0	2643	654	526	357	130	19	1	0	0	7	97	306	547	(8)	
	HDD18.3	4639	913	759	612	354	164	34	7	17	103	317	554	805	(9)	
	CDD10.0	1324	0	0	3	26	126	273	359	313	182	39	3	0	(10)	
	CDD18.3	279	0	0	0	1	13	57	108	71	28	2	0	0	(11)	
	CDH23.3	2794	0	0	1	16	186	573	1074	653	266	24	0	0	(12)	
	CDH26.7	846	0	0	0	3	57	169	364	179	69	3	0	0	(13)	
(14) Wind	WSAvg	3.7	3.9	3.9	4.0	4.3	4.0	3.5	3.0	2.9	3.4	3.9	3.9	3.8	(14)	
(15) Precipitation	PrecAvg	740	18	20	41	68	95	109	97	95	72	65	36	23	(15)	
	PrecMax	941	53	37	109	202	222	222	193	202	152	160	102	59	(16)	
	PrecMin	528	2	3	11	17	30	50	35	13	8	17	0	3	(17)	
	PrecStd	105	13	10	22	40	48	35	43	49	40	35	29	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	9.8	19.3	25.4	31.9	33.4	34.6	32.9	31.2	26.6	18.3	8.3	(19)	
		MCWB	2.7	6.6	13.1	15.1	19.6	23.2	24.5	23.7	21.4	17.7	12.3	5.3	(20)	
	2%	DB	2.7	5.6	14.7	21.8	28.0	30.6	32.1	30.5	28.7	22.5	14.2	4.7	(21)	
		MCWB	0.9	3.1	10.2	12.2	17.4	21.2	23.6	22.2	20.7	15.4	9.6	2.6	(22)	
	5%	DB	0.9	3.1	11.1	19.0	25.1	28.5	30.3	28.6	26.6	19.1	11.0	2.7	(23)	
		MCWB	-0.1	1.5	7.6	10.7	15.8	19.8	22.2	20.9	19.2	13.2	7.2	1.2	(24)	
	10%	DB	-0.7	1.3	7.8	16.3	22.4	26.7	28.6	27.0	24.1	16.7	8.4	1.1	(25)	
		MCWB	-1.7	0.0	4.9	9.4	14.5	18.8	21.1	20.2	18.2	12.1	5.6	0.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	6.9	15.2	16.4	21.9	24.8	26.7	25.5	23.8	19.2	13.7	5.8	(27)	
		MCDB	5.4	9.8	17.8	23.6	29.1	31.0	31.5	30.1	28.2	23.6	16.7	7.6	(28)	
	2%	WB	1.2	3.5	10.9	13.7	19.1	22.9	25.0	23.8	22.0	16.6	10.5	2.9	(29)	
		MCDB	2.5	5.3	14.0	19.6	24.4	28.6	30.2	28.5	26.6	20.7	13.7	4.4	(30)	
	5%	WB	0.0	1.6	7.8	11.7	17.4	21.5	23.7	22.6	20.7	14.5	7.7	1.4	(31)	
		MCDB	1.0	3.1	11.1	17.7	23.4	26.7	28.6	26.9	24.5	18.5	10.6	2.6	(32)	
	10%	WB	-1.7	0.1	5.1	10.0	15.7	20.2	22.5	21.3	19.2	12.4	5.7	0.1	(33)	
		MCDB	-0.7	1.4	7.9	15.6	21.0	24.8	27.2	25.2	23.1	15.9	8.2	1.1	(34)	
(35) Mean Daily Temperature Range	MDBR		9.4	10.1	9.9	11.6	12.4	11.7	12.1	12.0	12.1	10.7	9.0	8.4	(35)	
	5% DB	MCDBR	9.8	10.6	13.5	16.6	17.0	14.2	13.6	13.7	14.7	15.2	13.4	10.0	(36)	
		MCWBR	8.4	8.3	9.0	9.2	7.9	6.5	6.3	6.5	7.4	8.7	9.0	8.0	(37)	
	5% WB	MCDBR	9.7	9.7	12.7	14.3	13.8	12.0	12.1	11.6	12.2	12.8	12.3	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.263	0.276	0.312	0.369	0.402	0.391	0.385	0.399	0.367	0.314	0.285	0.260	(40)	
	taud		2.401	2.359	2.379	2.299	2.261	2.337	2.376	2.336	2.398	2.537	2.509	2.433	(41)	
	Ebn at Noon		859	913	922	889	867	876	877	850	850	861	830	826	(42)	
	Edh at Noon		71	90	103	122	132	123	117	117	101	76	64	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.71	2.80	3.84	4.68	5.37	5.96	6.34	5.29	4.06	2.55	1.62	1.19	(44)	
	RadStd		0.21	0.29	0.34	0.41	0.42	0.41	0.22	0.42	0.29	0.36	0.15	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page